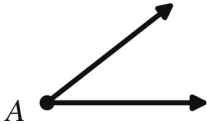
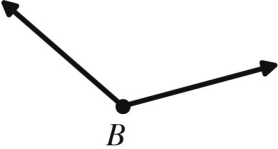
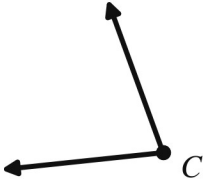
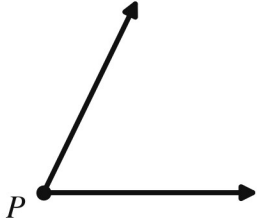


Part 1 — Copy each angle onto the ray below it. Compass and straightedge only. The marked point is the new vertex. Leave your arcs showing.

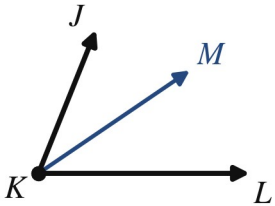
 A  A' m∠A = _____ m∠A' = _____ ★ Think deeper: Your second arc on the copy is a little too WIDE. Is the copy too big or too small? Why? _____
 B  B' m∠B = _____ m∠B' = _____ ★ Think deeper: Why MUST the copy be congruent? What shape are you really copying? (Hint: join the crossings.) _____
 C  C' m∠C = _____ m∠C' = _____ ★ Think deeper: If ∠C were drawn with sides 3 inches long, would m∠C change? Would your copy? _____

Part 2 — Bisect each angle. Compass and straightedge only. Then measure BOTH halves with a protractor.

 P m∠P = _____ half = _____ ★ Think deeper: Your two equal arcs cross at a SECOND point too. Would a bisector through THAT point work? Why? _____ _____
 Q m∠Q = _____ half = _____ ★ Think deeper: Keep bisecting one half. PREDICT how many bisections until the angle is under 10°, then check. _____ _____
 S R T ∠SRT is a straight angle. Bisect it. What kind of angle did you create? _____ ★ Think deeper: From a line, bisecting and copying only: which of 22.5°, 30°, 60°, 135° can you make? Why? _____ _____

Part 3 — Bisector and Angle Addition algebra. Write the equation, solve for x , then find EVERY angle asked for.

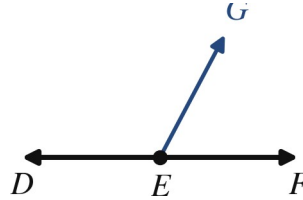
- a.** Ray KM bisects $\angle JKL$.
 $m\angle JKM = (4x + 5)^\circ$ and
 $m\angle MKL = (6x - 13)^\circ$. Find x
and $m\angle JKL$.



bisector $\Rightarrow$ halves EQUAL:
 $m\angle JKM = m\angle MKL$

★ **Think deeper:** Suppose the halves were $(4x + 5)^\circ$ and $(4x - 13)^\circ$ instead. Solve it. What happens, and what does it MEAN?

- b.** $\angle DEF$ is a straight angle.
 $m\angle DEG = (3x + 10)^\circ$ and
 $m\angle GEF = (2x - 5)^\circ$. Find x
and both angle measures.



straight angle $\Rightarrow$ parts ADD to 180°

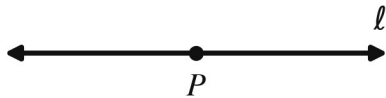
$$m\angle DEG + m\angle GEF = 180$$

★ **Think deeper:** Before solving, which values of x were ALREADY impossible? Use BOTH expressions.

- c.** Rey says bisecting an angle ALWAYS produces two acute angles. Is he right? Use your Part 2 results to explain.

★ **Think deeper:** Can an angle have TWO different bisectors? Use the Protractor Postulate to explain.

Part 4 — Team Challenge. Work with your group. Compass and straightedge only — the protractor is for CHECKING at the end.



Construct a 45° angle with its vertex at P — no protractor allowed.

★ **Think deeper:** You now own 45° . Describe how to build 67.5° next — still no protractor.

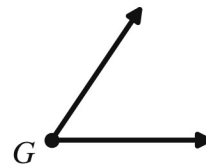
- a.** Plan first: which two constructions from this lab take you from 180° down to 45° ?

- b.** Build it on the figure, then check with a protractor.

$$m\angle = \underline{\hspace{2cm}}$$

- c.** Why is it EXACTLY 45° even before you measure?

- d.** Extension: on the ray from V , construct an angle whose measure is $1\frac{1}{2} \cdot m\angle G$. Write your plan before you build.



Check: $m\angle V = \underline{\hspace{2cm}}$